Junghoon Chung

540-808-5715 — junghoon@uw.edu — www.linkedin.com/in/junghoon-chung — sites.google.com/uw.edu/junghoonchung

Summary

PhD candidate in Industrial and Systems Engineering specializing in **machine learning, causal inference, and multimodal human behavior modeling**. Experienced in developing **end-to-end pipelines and evaluation frameworks** that translate complex data into actionable, user-centered insights. Skilled in applying **AI/ML methods, statistical analysis, and experimental design** to real-world problems, with publications in human factors and ergonomics. Seeking a research or data science internship to develop adaptive, data-driven solutions.

Education

University of Washington, Seattle — **Sep 2025 – Jun 2029 (Expected)**
Ph.D in Industrial and Systems Engineering, Dean's Fellowship Recipient

Virginia Tech, Blacksburg — **Aug 2023 – May 2025**
M.S in Industrial and Systems Engineering (Thesis Track), GPA 3.89/4.0

Yonsei University, Seoul — **Mar 2017 – Feb 2023**
B.S in Industrial Engineering, GPA 3.52/4.0, Student Council President

Experience

Graduate Research Assistant, ***Human Automation Systems Lab, University of Washington*** — **Sep 2025 – Present**
- Built machine learning models to assess vigilance using multimodal physiological time-series data (ECG, eye tracking, BVP, EDA, HR). Developed preprocessing pipelines for heterogeneous sensor streams using Python and scikit-learn
- Engineered 30+ physiological and contextual features across over 10,000 observations per participant
- Improved predictive performance by 20% through feature engineering and model refinement

Graduate Research Assistant, ***Occupational Ergonomics and Biomechanics Lab, Virginia Tech*** — **Aug 2023 – May 2025**
- Developed end-to-end data pipeline to collect, preprocess, and analyze multimodal datasets including EMG, EEG, IMU, motion capture, and behavioral logs
- Processed longitudinal datasets (3 weeks per participant; over 100,000 records) for modeling posture and behavioral transitions. Designed and validated zero-shot NLP classification model to categorize unstructured work-context text
- Built time-series forecasting models incorporating contextual variables. Improved prediction accuracy through data segmentation and clustering of user behavior patterns
- Identified drivers of behavioral outcomes and communicated findings through technical reports and data visualizations

Key Projects

• **Human Vigilance Modeling with causal-inference models, — University of Washington**
Built predictive models of moment-to-moment alertness using **multimodal physiological and contextual data, engineering 30+ features across over 10,000 observations** per participant. Improved model performance by 20% through feature engineering and tuning, and applied causal inference to identify key drivers of performance variability.

•**Context-Aware Sit-Stand Intervention for Promoting Healthy Behaviors in Knowledge Workers, — Virginia Tech**
Analyzed **longitudinal behavioral datasets (3 weeks per participant; over 100,000 data points per participant) to forecast posture and behavioral transitions**. Developed **time-series models incorporating contextual variables** and improved prediction accuracy through **data segmentation and clustering-based personalization**.

•**Review reliability analysis by BERT sentiment analysis and Clustering — Yonsei University**
Analyzed over **50,000 online product reviews using transformer-based sentiment modeling and clustering** to identify bias patterns across reviewer groups. Proposed **cluster-normalized scoring methods to reduce user bias and improve the reliability** of aggregate product ratings derived from large-scale user-generated data.

Skills

Programming: Python, SQL (MySQL)
ML: Scikit-learn, TensorFlow, PyTorch
Data Analysis: Statistical modeling, Experimental design, Time-series analysis
Communication: Technical reporting, Cross-functional collaboration

Awards

Dean's Fellowship, College of Engineering — **University of Washington, 2025 –2029**
Social-Innovation Capability Video Contest 2nd Place – Data mining — **Yonsei University, 2022**
Merit-based Scholarship (Academic Excellence) — **Yonsei University, 2021**